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Project: Inferential Statistics; Hypothesis Testing

Topic 1: Inferential Statistics: Population Mean: One Sample

Dataset: Female Pulse Rates (bpm) Beats per Minute

Source:

<https://www.statcrunch.com/app/?dlim=comma&ft=true&dataurl=https%3a%2f%2fidx.acs.pearson.com%2fGetPlayerFile.ashx%3fguid%3ded789ac2-8690-4aa8-b3ed-4c4c67cfaba5#>

Description: The dataset is the pulse rates of 40 adult females.

Row	Females
1	81
2	96
3	58
4	67
5	54
6	80
7	79
8	88
9	86
10	57
11	38
12	67
13	83
14	77
15	76
16	64
17	67
18	80
19	60
20	66
21	86
22	79
23	68
24	73
25	86
26	87
27	89
28	93
29	91
30	95

Row	Females
31	68
32	89
33	80
34	79
35	75
36	57
37	102
38	75
39	75
40	76

Sample size, $n = 40$

Objectives:

- (1.) Construct a 95% confidence interval estimate of the population mean of pulse rates of adult females in beats per minutes.
- (2.) Interpret the confidence interval.

Parameter to estimate: Population Mean

Test: t-test

Verify Requirements for the Test.

- (1.) The sample is a simple random sample.

- (2.) The sample size is at least 30. This satisfies the Central Limit Theorem of normal distribution requirement.

Data Analysis Using R Language

```
# Load the dataset
data <- read.csv("PulseRatesFemales.csv")

# 95% Confidence Interval for Females
females_ci <- t.test(data$Females, conf.level = 0.95)$conf.int
print(paste("Females 95% CI:", females_ci[1], "to", females_ci[2]))
```

```
[1] "Females 95% CI: 71.933473728821 to 80.416526271179"
```

We are 95% confident that the population mean is approximately between 71.9 bpm and 80.4 bpm. (Rounded to one decimal place.)

Topic 2: Hypothesis Testing: Testing a claim About Population Mean

Dataset: Volumes of Regular Coke Cans (in Ounces)

Source:

<https://www.statcrunch.com/app/?dlim=comma&ft=false&dataurl=https%3a%2f%2ftdx.acs.pearson.com%2fGetPlayerFile.ashx%3fguid%3d4b1e424a-a1a4-41b2-b897-5d16b6c9fbb9>

Description: The dataset I shall use is the Volumes in ounces of 36 cans of regular Coke.

Row	RegularCokeCanVolumes
1	12.32
2	12.12
3	12.24
4	12.32
5	12.23
6	12.27
7	11.98
8	12.14
9	12.24
10	12.09
11	12.32
12	12.28
13	11.83
14	12.26
15	12.11
16	12.12
17	12
18	12.2
19	12.16
20	12.15

21	12.23
22	12.22
23	12.24
24	12.35
25	12.17
26	12.21
27	12.34
28	12.23
29	12.21
30	12.33
31	12.17
32	12.19
33	12.06
34	12.36
35	12.16
36	12.18

Variable: Volumes of regular Coke cans in ounces

The accompanying data set includes volumes (ounces) of a sample of cans of regular Coke.

The summary statistics are

Sample size, $n = 36$

Sample mean, $\bar{x} = 12.195$ oz

Sample standard deviation, $s = 0.112$ oz

Assume that a simple random sample has been selected. Use a 0.05 significance level to test the claim that cans of Coke have a mean volume of 12.00 ounces.

Objectives:

- (1.) Test the claim that the mean volume of Coke cans is 12.00 ounces using the Critical Value Method.
- (2.) Test the claim that mean volume of Coke cans is 12.00 ounces using the Probability Value Method.
- (3.) State the decision.
- (4.) State the conclusion.
- (5.) State the interpretation.

Verify Requirements for the Test.

- (1.) The sample is a simple random sample.
- (2.) The sample size is at least 30. This meets the normality requirements based on the Central Limit Theorem.

Null Hypothesis: $\mu = 12 \text{ ounces}$

Alternative Hypothesis: $\mu \neq 12 \text{ ounces}$

Test: One Sample two-tailed

Data Analysis using R Language

```
# Load the data
data <- read.csv("RegularCokeCanVolumes.csv")
volumes <- data[[1]] # Select the first column

# Perform the t-test
testResults <- t.test(volumes, mu = 12.00, conf.level = 0.95)

# Display results
print(testResults)

# Calculate Critical Value for Method 1
alpha <- 0.05
degreesFreedom <- length(volumes) - 1
tCritical <- qt(1 - alpha/2, degreesFreedom)
cat("Critical Value (t): +/-", tCritical, "\n")
```

One Sample t-test

```
data:  volumes
t = 10.456, df = 35, p-value = 2.607e-12
alternative hypothesis: true mean is not equal to 12
95 percent confidence interval:
 12.15736 12.23319
sample estimates:
mean of x
 12.19528

Critical Value (t): +/- 2.030108
```

Critical Value Method:

$$\text{Critical } t - \text{value} = \pm 2.030108$$

$$\text{Test statistic} = 10.456$$

$$10.456 > 2.030108$$

The test statistic is greater than the critical t-value. This means that it falls in the critical region.

Decision: Reject the null hypothesis.

Probability Value Method:

$$\text{probability value} = 0$$

$$\text{significance level} = 0.05$$

$$0 < 0.05$$

The probability value is less than the significance level.

Decision: Reject the null hypothesis.

Conclusion: There is sufficient evidence to support the claim that the cans of Coke do not have a mean volume of 12.00 ounces.

Interpretation: The population mean of the volumes of coke cans is significantly different than 12 ounces.

References (MLA 9)

“StatCrunch.” *Statcrunch.com*, 2026,

www.statcrunch.com/app/?dlim=comma&ft=true&dataurl=https%3a%2f%2f2ftdx.acs.pearson.com%2fGetPlayerFile.ashx%3fguid%3ded789ac2-8690-4aa8-b3ed-4c4c67cfaba5#. Accessed 29 Apr. 2026.

“StatCrunch.” *Statcrunch.com*, 2026,

www.statcrunch.com/app/?dlim=comma&ft=false&dataurl=https%3a%2f%2f2ftdx.acs.pearson.com%2fGetPlayerFile.ashx%3fguid%3d4b1e424a-a1a4-41b2-b897-5d16b6c9fbb9. Accessed 29 Apr. 2026.